

Meeting Report: YZV202E Project Update

Scope Revision and Task Allocation

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1. Project Scope Revision

Following our recent discussion, the scope of the project has been officially revised. Instead of optimizing the entire megacity public transportation network, the problem space has been narrowed to focus exclusively on the **Metro lines of Istanbul**. This reduction in scope will allow for a more precise, computationally feasible simulation and optimization process.

2. Methodology Updates and Task Sharing

2.1. Data Cleaning and Aggregation (Assigned to: Huzeyfe)

The current dataset contains granular, row-by-row ticket usage data, which includes unnecessary variables such as usages that are out of our scope and unnecessary details about the usage. This data will be cleaned and aggregated into a streamlined format containing exactly four columns: **Metro Line, Station, 1-Hour Time Interval, and Number of Tickets Used**.

Since exact ticket timestamps are unavailable, we will assume a uniform distribution of passenger arrivals across the hour, resulting in a constant minute-by-minute boarding frequency. Additionally, the mean hourly usage will be calculated on a monthly basis to account for macro-level seasonal variations. Establishing this baseline demand is necessary to accurately measure the impact of external events on the network.

2.2. Event Data Collection (Assigned to: Emir)

To evaluate the responsiveness of the system to unpredictable demand anomalies, the scope of public events has been restricted to major football matches. Specifically, data collection will focus on the home matches of Istanbul's 3 big clubs: Galatasaray, Fenerbahçe, and Beşiktaş. The dataset will catalog the exact dates and times these matches occur at their respective home stadiums.

2.3. Operational Cost Formulation (Assigned to: Erdem)

A dedicated function will be developed to accurately calculate the operational cost, represented by the β_k variable in the objective function. For this model, the vehicle type index (k) will be simplified to two distinct configurations:

- $k = 1$: Full capacity (e.g., an 8-wagon train).

- $k = 2$: Half capacity (e.g., a 4-wagon train).

The formulation for β_k will incorporate the logistical cost of assigning a driver (makinist) to a specific train at a given time, as well as the mechanical power and fuel consumption costs associated with running that specific vehicle capacity.

2.4. Directional Passenger Distribution Coefficients (Assigned to: Atila)

Upon reviewing the available dataset, it was identified that destination data for travelers is missing; the records only contain the origin station, line, and time interval. To address this, a proportional distribution model will be implemented to divide boarding passengers between the two travel directions based on the relative utilization of the stations in each direction.

For example, if a line has 11 equally used stations and 100 passengers board at the 2nd station, 90% of them will be assumed to travel toward the direction with 9 stations, and 10% toward the direction with 1 station. Furthermore, this split will be weighted by the actual historical usage rates of those target stations. If the 1st station accounts for 20% of the line's total ticket usage, it will proportionally attract 20% of the traffic from the 2nd station. Atila will be responsible for calculating these utilization coefficients for every station across all metro lines to accurately estimate passenger flow directions.

2.5. Demand Anomaly Parameterization (General Team Task)

This task will also be done collectively by the team. To enhance the realism of the model, specific load parameters will be calculated to determine how much passenger volume increases at individual stations during the aforementioned football events. This involves quantifying the varying impact of the three major events on *every* station in the network, which will directly inform the event intensity factor E_t and the adjusted passenger demand equation:

$$\tilde{D}_{s,t} = D_{s,t}(1 + \delta_1 W_t + \delta_2 E_t)$$